

SUBHAM TOMAR

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PROFESSIONAL SUMMARY

Final-year B.Tech (Information Technology) student and Generative AI enthusiast with hands-on experience in Python and backend development. Built AI-powered conversational systems and machine learning applications using FastAPI, REST APIs, and full-stack technologies. Seeking a Software Development, Backend, or Generative AI internship.

EDUCATION

Pranveer Singh Institute of Technology, Kanpur

B.Tech in Information Technology | CGPA: 7.2 | Expected Graduation: 2027

Class XII (CISCE/ISC) — 70%, 2023

Class X (CISCE/ICSE) — 80%, 2021

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI / ML: Numpy, Pandas, Machine Learning, Scikit-learn, TensorFlow, Generative AI

Backend: FastAPI, REST APIs

Databases: PostgreSQL

Core CS: Data Structures & Algorithms, Software Engineering, DBMS

Tools: Git, GitHub, VS Code

INTERNSHIP EXPERIENCE

Python Developer - QSkill

July 2026 – Present

- Building an AI-powered conversational assistant in Python using the Google Gemini API, implementing conversation memory to retain prior user interactions and generate context-aware responses.
- Integrating the Google Search API to surface real-time information, including weather updates, cryptocurrency prices, and current news.
- Architecting a modular backend for prompt handling, API integration, and conversation management, with emphasis on clean code, debugging, and performance optimization.

PROJECTS

NetScope – AI-Powered Deep Packet Inspection Engine [\[Github\]](#)

Jan 2026 – Mar 2026

- Engineered an AI-powered Deep Packet Inspection (DPI) engine achieving 99.2% traffic classification accuracy with <1.8 ms per-packet processing latency.
- Reduced CPU processing overhead by 87% using an O(1) Fast-Path TLS/SNI parser before ML-based traffic classification.
- Designed a multithreaded packet-processing pipeline supporting throughput of 10,000+ packets per second (PPS).

AegisCRM-AI-First-HCP-Interaction-Management [\[Github\]](#)

Apr 2025 – Jul 2025

- Engineered a modular FastAPI backend with 6 REST and AI chat API endpoints, leveraging SQLAlchemy ORM to manage interactions in a structured SQLite database.
- Developed a stateful AI agent using LangGraph and Groq API, integrating 5 custom database-aware tools for automated entity extraction, follow-up recommendations, and record management.
- Implemented Pydantic validation schemas to ensure type-safe request handling and reliable data validation across API interactions.

Next Word Prediction using LSTM [\[Github\]](#)

Mar 2024 – Apr 2024

- Designed and built a lightweight LSTM-based text prediction system, reducing model footprint by 98% (862 KB) compared to standard Transformer models.
- Engineered an asynchronous FastAPI backend using Uvicorn and Pydantic, enabling efficient in-memory inference for next-word prediction.
- Developed automated Pytest test cases covering all backend API routes to validate prediction and request handling workflows.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle | Issued Aug 2025 · Expires Aug 2027 | Credential ID: OC6909716
- Google AI Essentials — Coursera | Issued Jun 2026 · Expires Jun 2028 | Credential ID: C66GCCRRRL4AL
- Google Prompting Essentials — Coursera | Issued Jun 2026 · Expires Jun 2028 | Credential ID: O44GJHI3B1LW

ACHIEVEMENTS

- Awarded Silver Medal at the District-Level Taekwondo Championship.
- Recognized for outstanding performance in sports activities during NCC camps.